HPGe phonon propagation

Reusable research system and campaign specification | Athul Prem | Revised 12 November 2026

Develop a reusable three-dimensional elastic-wave simulation system for nominally pure high-purity germanium (HPGe). Use it to measure how geometry, thickness and temperature affect phonon-mode propagation, reflection, mode conversion and localization. The mandatory thicknesses are 100 nm, 200 nm and 1 micrometre. Repeat identical reference geometries at each temperature, then compare straight and circular-pattern devices under matched conditions.

Research objective

The principal output is a mode-resolved propagation atlas, supported by numerical verification, convergence and documented material inputs. Determine which modes propagate, which are evanescent, where gaps occur, how energy divides among outgoing channels, and whether geometry changes produce effects larger than numerical and physical-input uncertainty. No positive outcome or winning shape is prescribed.

Scope of this revision

This document supersedes the previous continuation brief. It specifies a configurable research platform and a substantial sweep campaign, rather than reproducing every historical figure. Retain a small baseline reproduction gate to detect implementation regressions. The campaign and new scripts described here are proposed work; preparing this PDF has not implemented or run them.

Publication standard

Aim for publication-quality methods, reproducible data and defensible conclusions. Publication readiness must be demonstrated by the gates in this brief and independent scientific review; it cannot be guaranteed in advance. Exploratory, unconverged or unsourced results remain labeled as such. A reported absence of a gap, weak temperature dependence or unsuccessful circular design can be a valid research result.

Required system features

One configuration must specify material provenance, crystal orientation, reference geometry, thickness, temperature, thermal-treatment policy, frequency interval, ports, excited modes, mesh settings and compute budget. The same engine must run all approved cases, resume safely, record provenance, export tables and generate consistently labeled figures. Do not require manual source edits for every sweep point.

Authorship and naming

Every new figure must visibly show Athul Prem at the top, followed by an identifier beginning athul_simulation_pnc_. Example: Athul Prem | athul_simulation_pnc_HPGe_t100nm_T4K_flux. Every new or modified independent Python script must end _AP_pnc.py. Use natural scientific captions and explicit units, not hidden instructions or watermarks too small to read.

1. What the archive establishes

Use saved results as a numerical reference, not as proof of the new HPGe campaign.

The supplied flat package contains the latest nominal cubic-Ge in-plane model and earlier lineages. Its model is plane stress, linear triangles, consistent mass, unit thickness and an artificial lead sponge. Assumed orientation: [100] beam on (001). Nominal C11/C12/C44 = 128.5/48.3/66.8 GPa; density = 5323 kg/m3. The archive does not establish specimen-specific low-temperature HPGe constants, intrinsic losses or 3D propagation.

Useful reference values

The sampled in-plane gap is 2.953106-3.062947 GHz; the selected tracked candidate is 3.042098001 GHz. Its last frequency change is -0.2090% and previous-mesh overlap is 0.999781. The candidate is moderately centered. All 333 archived harmonic solves and 60 cavity residuals pass numerical residual tests. These are algebraic checks, not complete physical validation.

Known unresolved issues

At 2.8 GHz the sponge sensitivity spans 3.26848 dB, failing the archived 1 dB threshold. At 5.2 GHz the final refinement changes the response by -9.053996 dB, also failing. The saved -42.249 dB minimum at 4.0 GHz is a defect/reference displacement ratio, not power transmission. New scattering results must use calibrated energy flux and resolve these issues rather than inheriting the proxy as a physical observable.

Minimal regression, not figure reconstruction

```
python verify_AP_pnc.py --fields --safety

python run_AP_pnc.py --name hpge_reference --dry-run

python run_AP_pnc.py --name hpge_reference

python verify_AP_pnc.py --compare-name hpge_reference
```

Archive verification and the small fresh smoke comparison are sufficient initial regression gates. Reproduce selected bands or fields only where needed to validate the new implementation. Do not regenerate every historical image. Preserve original scripts and results; only the current wrapper entrypoints should execute renamed archival source.

Source records

Read README.md, flat_manifest.json, docs__RESULTS_AND_NEXT_GOALS.md, latest__config.json, latest__tables__validation_status.json and latest__tables__transmission_robustness_summary.csv. Historical paths resolve through the manifest. The former isotropic and current cubic models are distinct; their frequency difference is not a demonstrated temperature shift. New campaign data must have separate run IDs and model versions.

2. Define pure HPGe and temperature

Work package A | Establish a material model before interpreting thermal effects.

A1. Purity is a physical input

Use crystalline HPGe without intentional doping or added layers. Record actual impurity concentration or resistivity where available, isotopic composition, defect/dislocation information, crystal axes and specimen temperature. Chemical purity does not imply isotopically enriched Ge or eliminate boundary scattering. If no specimen certificate is available, label the model nominal pure HPGe and retain material uncertainty rather than claiming measured purity.

A2. Temperature grid

Proposed initial grid: 0.02, 0.04, 1, 4, 20, 77 and 300 K. These are campaign design points, not supplied experimental conditions. Use all three required thicknesses at each admitted temperature. Do not extrapolate low-temperature elastic or damping data across this range without a justified model. A point lacking supported properties may be computed only as a clearly labeled fixed-property control or sensitivity scenario.

A3. Material registry

For each material parameter save value, units, temperature interval, source, original measurement context, uncertainty and interpolation rule. Required entries are C11(T), C12(T), C44(T), density rho(T), thermal expansion alpha(T), and any loss/relaxation model. Read primary references and errata. Verify stiffness positivity and tensor symmetry throughout interpolation. Do not fit arbitrary losses to make propagation decay or resonances look realistic.

A4. Separate three temperature questions

Control I: freeze geometry, tensor, density and damping while changing Bose occupation. Classical normalized transmission and eigenfrequencies must remain identical; only occupation-weighted quantities change. Model II: keep geometry fixed while varying supported material properties. Model III: allow free thermal contraction or a separately modeled constrained thermal-stress state. These answer different questions and must not be merged into one unexplained temperature curve.

For free isotropic contraction, use a documented scale factor obtained from the expansion law and maintain mass consistency between density and volume. For constrained supports solve thermoelastic equilibrium first and include a justified prestress linearization. Define the reference temperature and distinguish as-fabricated dimensions from dimensions at measurement temperature.

Acceptance

Deliver material_provenance.csv, material_registry.json and uncertainty curves. A finite temperature effect on normalized propagation must identify its physical cause. If only Bose weighting changes, state that no temperature-dependent mechanical propagation has been established.

3. Geometry and thickness matrix

Work package B | Construct comparable physical devices, not thickness-scaled 2D plots.

B1. Mandatory dimensions

Use t = 100 nm, 200 nm and 1000 nm. Hold the same in-plane geometry and material orientation fixed across this thickness comparison. The existing 26-hole workbook supplies a baseline hole pattern; its beam width of 1000 nm and leads of 1500 nm remain provisional. Preserve these values for the initial controlled comparison, then study dimension sensitivity separately. Explicitly record all hole coordinates, diameters, taper lengths, ligament widths and lead dimensions.

B2. Why 3D is necessary

Build real extruded 3D solids with free top, bottom and hole surfaces, except explicitly modeled supports or ports. Changing a plane-stress model's scalar thickness does not create thickness modes, flexural/torsional dispersion or out-of-plane leakage. A 1 micrometre-thick beam of 1 micrometre width is not a thin membrane. Use 2D only as a labeled screening approximation and benchmark, never as evidence for thickness-dependent full elastic transmission.

B3. Reference and comparison devices

For each thickness create a homogeneous beam, periodic perforated waveguide and finite defect nanobeam. Add the circular branch only after its construction is validated. Lead cross-sections must match their attached device and the reference used for normalization. In-plane geometry comparisons must preserve thickness; thickness comparisons must preserve the selected in-plane geometry. Do not change both silently.

B4. Additional dimensions

After screening, vary width, hole diameter, pitch and defect taper around the baseline. Proposed exploratory factors are 0.9, 1.0 and 1.1 of a reference dimension, subject to positive ligament and mesh-quality constraints. These are sensitivity bounds, not asserted fabrication tolerances. Begin with one-variable studies, then a small interaction design for parameters that matter. Report physical dimensions, not only dimensionless scale factors.

Geometry acceptance and outputs

Check closed volumes, non-intersection, minimum separation, consistent physical groups and correct port normals. Export dimensioned top/side views, a complete parameter table and geometry checksums. Build at least two independent mesh realizations for a representative shape. Preserve exact geometry under refinement rather than altering hole contours unintentionally. Stop infeasible cases before meshing and record why they were rejected.

4. Reusable simulation architecture

Work package C | Implement once, then run controlled campaigns.

Modules to develop

material_registry_AP_pnc.py: property lookup, tensor rotation, temperature policy and uncertainty. geometry_factory_AP_pnc.py: straight, bent and ring construction plus validity checks. mesh_controls_AP_pnc.py: wavelength/curvature/thickness refinement. lead_modes_AP_pnc.py: channel calculation and power normalization. solve_scattering_AP_pnc.py: driven elastic fields and scattering coefficients. sweep_campaign_AP_pnc.py: dependency-aware execution, checkpoints and resume. analyze_results_AP_pnc.py and plot_results_AP_pnc.py: metrics, uncertainty, tables and labeled figures. validate_physics_AP_pnc.py: verification and regression suite. These are proposed new files, not current package commands.

Configuration contract

Store material_id; material_model_version; crystal_rotation; geometry_family; geometry_parameters_SI; thickness_m; temperature_K; reference_temperature_K; thermal_policy; frequency_min_Hz; frequency_max_Hz; frequency_strategy; incident_mode_ids; port_definitions; mesh_policy; solver_tolerances; boundary_model; uncertainty_case; and resource_limits. Reject ambiguous units, missing provenance and incompatible geometry/port choices. Save the fully resolved configuration with every result.

Execution and restart

Hash configuration, source and inputs into a run identity. A change in any scientific input creates a new run. Save completed frequency/mode tasks atomically; reuse only matching hashes. Use locks against simultaneous writes. Failed points keep diagnostics and are not filled by interpolation in final tables. A resumed case must demonstrate identical results to a clean small run.

Result schema

Each row identifies geometry, thickness, temperature, material scenario, frequency, incident/outgoing channel, polarization, propagation constant, group velocity or power direction, T, R, absorption, energy imbalance, mesh DOFs, solver residual, convergence state and uncertainty. Store absent channels separately from failed computations. Distinguish real propagating modes and evanescent modes used for boundary matching.

Portable flat handoff

Keep the shared handoff root flat if that remains the chosen format, using deterministic prefixes and a machine-readable index. Solver scratch trees and virtual environments stay outside it. Preserve the existing adapter and archived evidence. Document exact new entrypoints, dependencies and example configurations; test a new installation on the colleague's machine before a large sweep.

5. Which modes and how many

Work package D | Enumerate the physics, not an arbitrary number of eigenvalues.

D1. Mode families

Resolve longitudinal/extensional, in-plane flexural, out-of-plane flexural, torsional and mixed modes, plus higher-order transverse/thickness branches. Labels must come from displacement and strain-energy projections, parity where applicable and continuity along the dispersion curve. In anisotropic or curved structures modes may be hybridized; report mixed character instead of forcing every branch into a pure label.

D2. Count all open channels

At each frequency solve the lead eigenproblem sufficiently completely to identify every propagating channel within the study window. Use power flow to classify incoming/outgoing directions and treat cutoff modes carefully. Increase the requested eigenpair count or contour-search coverage until no additional open channels appear. The maximum mode count depends on thickness, frequency and lead cross-section; it is not a fixed universal number.

D3. Evanescent content and mode tracking

Retain enough evanescent modes or lead-domain length for scattering results to converge. Near degeneracies use subspace tracking and energy/power orthogonality, not nearest frequency alone. Record cutoff frequencies and mode conversion. For non-Hermitian lead formulations use the appropriate left/right modal normalization rather than assuming ordinary Euclidean orthogonality.

D4. Quantities per channel

Export dispersion, group velocity or independently computed energy velocity, polarization fractions, power normalization, mode profile and cutoff status. Compute a scattering matrix linking each excited incoming channel to every outgoing open channel. Present both channel-resolved transmission and total transmitted power. A displacement probe can supplement these quantities but cannot replace them.

Tests and products

Verify homogeneous-beam wave speeds against the anisotropic Christoffel calculation where applicable; verify long-wavelength rod/beam limits in their regime. Show modal completeness versus search size, representative profiles for each family and branch tracking through crossings. Provide mode_inventory.csv and scattering_channels.csv. For the 1 micrometre case expect more channels, but determine the actual count numerically rather than asserting a count in advance.

Acceptance: increasing mode-search size and evanescent content changes resolved transmission by less than the declared convergence tolerance. No unexplained loss of power is allowed simply because a converted outgoing mode was omitted.

6. Frequency sweep and propagation

Work package E | Broad coverage with adaptive resolution and defensible power accounting.

E1. Frequency range and practical maximum

Proposed primary interval: 0.1-6 GHz, covering the saved near-3 GHz region and the problematic 5.2 GHz point. Establish supported material and spatial-resolution validity before extending it. Use 8 GHz only as an optional extension after convergence and resource review. Do not include exactly zero frequency in a harmonic scattering sweep. There is no meaningful universal maximum number of frequencies: the required count follows resonances, cutoffs and uncertainty.

E2. Adaptive sampling

Start each representative spectrum with 121 points across the primary interval; insert cutoffs, known feature neighborhoods and validation frequencies explicitly. Refine where T/R changes rapidly, phase changes rapidly, interpolation error is excessive or a resonance is detected. Use an initial planning cap of 1001 points per spectrum, not a scientific completeness guarantee. If unresolved features remain at the cap, flag the case and request a revised budget. Resolve isolated linewidths with at least 10-20 points across the linewidth and demonstrate stability under further refinement.

E3. Energy transport

With exp(-i*omega*t), v = -i*omega*u. Use time-averaged elastic flux S_j = -0.5 Re(sum_i stress_ij*v_i*) and verify the sign on a known traveling wave. Integrate over the physical 3D port area. Normalize each incident mode to unit power; compute T and R from outgoing modal powers. Track 1-R-T-A. Sponge absorption is a numerical boundary treatment, not physical intrinsic absorption.

E4. Controls and propagation products

A matched homogeneous lossless beam should transmit approximately unit power. Check reciprocity using correct reciprocal channel pairs and power normalization. Move monitors and vary lead length, absorber length/profile and mesh independently. Plot displacement, strain energy, flux streamlines and channel conversion at representative passband, gap and resonance frequencies, not at every sampled point.

Acceptance

Proposed targets: power-budget error below 1% of incident power; representative T/R changes below 0.01 absolute under independent mesh, boundary and modal refinements. Tighten these if claimed improvements are smaller. Near a deep minimum use absolute errors and a numerical floor, not unstable relative errors. Never report a suppression level below the validated numerical floor as a measured value.

7. Temperature-dependent propagation

Work package F | Separate thermal occupancy from changes in the mechanical device.

F1. Matched sweep matrix

Run the same reference geometry at every admitted temperature and all three thicknesses. For each case state whether geometry is frozen, freely contracted or constrained. Produce T_mode(f,T,t), R_mode(f,T,t), cutoff trajectories, group/energy velocities, band-edge shifts and mode-conversion maps. Preserve matched frequency axes and polarization definitions. Track modes across temperature by field/subspace overlap rather than mode index.

F2. Fixed-property null experiment

Repeat at least two temperatures with identical mechanical inputs. The normalized scattering matrix must agree within solver error. Compute Bose occupation separately, n(f,T) = 1/[exp(h*f/(kB*T))-1], using numerically stable evaluation. Any claimed mechanical temperature dependence in this null experiment is a bug or a hidden parameter change.

F3. Loss and attenuation

A lossless elastic model can establish reflection, conversion, confinement and leakage, but not intrinsic phonon lifetimes. Introduce anharmonic, isotope, defect or surface scattering only through a physically sourced model with stated frequency/temperature validity. Do not assume ballistic transport over all lengths and temperatures. Compare device length with justified mean-free-path estimates and distinguish coherent elastic scattering from incoherent transport when the regime changes.

If using a complex stiffness or relaxation model, verify passivity, the time-convention sign and consistency with attenuation benchmarks. Separate amplitude attenuation from power attenuation; state the fitted length interval. Extract attenuation from matched devices of several lengths to separate entrance reflection from bulk loss. Include confidence intervals and avoid fitting through noise floors.

F4. Thermal transport, conditional extension

If a ballistic or suitably justified transmission model is available, compute occupation-weighted spectral power or linear-response conductance from the channel transmission and dn/dT. Use the appropriate integral and units. A 0.1-6 GHz spectrum does not capture total thermal conductance at every temperature, especially high temperatures; report a band-limited contribution unless the omitted range is demonstrably negligible. Do not call it bulk thermal conductivity without geometry, transport-regime and length justification.

Outputs

Deliver temperature-thickness heat maps, mode-resolved propagation spectra, material-versus-occupancy decomposition, attenuation-length studies where supported, and a regime/validity table. Negative thermal shifts or negligible changes are retained as scientific outcomes.

8. Circular pattern and fair comparison

Work package G | Compare curved propagation with the straight reference.

G1. Proposed interpretation

The straight archive already contains round holes. Implement circular pattern as holes arranged along a curved centerline, extending to an annular/ring structure. This is a declared design proposal, not a claim that the exact circular device has already been specified or simulated. Begin with an open bent beam connected to matched straight leads; then study a closed ring and a ring coupled to bus waveguides as separate topologies.

G2. Parameterization

Specify centerline radius R, width w, thickness t, angular positions, local hole radii, arc pitch, taper, hole count and coupling/support dimensions. Map local distance by s = R*theta. Check inner-edge clearance and hole overlap. For a full ring enforce circumference closure and explicitly report whether pitch, N or R is adjusted. A ring with a repeated identical unit has a different defect structure from a tapered cavity; label both accurately.

G3. Curvature sweep

Proposed open-bend screening radii are R/w = 5, 10, 20 and 40, plus a straight control; exclude infeasible shapes. Match path length and local pattern where possible, and disclose residual differences. Use t = 100, 200 and 1000 nm for selected feasible radii. Start at a well-supported reference temperature, then carry finalists through the same temperature grid. A full Cartesian campaign is not required before screening establishes useful designs.

G4. Crystal orientation and modes

Keep the single-crystal tensor fixed in laboratory coordinates while the tangent rotates. Rotating the crystal tensor with the local tangent describes a different material and is only a labeled diagnostic. Study curvature-induced polarization mixing, reflection, ring doublet splitting and bend leakage. Do not transplant a straight-cell gap into the ring without verifying its modes and propagation.

G5. Comparisons

For open devices compare calibrated T/R, conversion and localization at equal thickness, path length, local dimensions and material policy. For closed rings report eigenmode families; for coupled rings report through/drop scattering and coupling-dependent linewidth. Do not compare a closed ring's real eigenfrequency calculation to an open beam's radiative Q. Use the large-radius limit as a consistency test and report trade-offs rather than forcing one shape to win.

9. Numerical and physical validation

Work package H | A result qualifies for publication claims only after its relevant gates pass.

Verification gates

Test stiffness and mass symmetry, rigid-body modes, uniform-strain patch behavior, units, tensor rotations and passive energy balance. On small identical problems compare independent eigensolvers and, where practical, an independent implementation. Retain dimensionless algebraic residuals and mode orthogonality. Proposed small-benchmark residual target: 1e-8 or better, with conditioning and attainable accuracy documented.

Spatial and spectral convergence

Use at least three systematically refined meshes for representative cases of every thickness and geometry family. Refine shortest wavelength, through-thickness structure and curved hole surfaces independently. No fixed elements-per-wavelength rule replaces convergence. Report DOFs, element order, quality, wall time and memory. Refine wavevector sampling separately from mesh in periodic problems. Resolve resonance linewidth and mode completeness independently from spatial refinement.

Proposed initial acceptance tolerances

Tracked eigenfrequency and gap-edge change: below 0.5% over the last refinement, with a stable multi-level trend. Resolved non-degenerate mode overlap: above 0.99, or stable subspace overlap for degeneracies. Localization-fraction change: below 0.02 absolute. Calibrated power errors and sensitivity: below 0.01 absolute. These are starting targets, not achieved facts; tighten them when uncertainty could change a conclusion. Q and attenuation require independent convergence and fitted uncertainty.

Physical validity gates

Require sourced material data, explicit purity/orientation, real 3D thickness geometry, matched ports, valid temperature/loss regime and a complete accounting of outgoing power. Quantify uncertainty from material properties separately from discretization. Compare with an experimental or trusted external benchmark when available; otherwise call the work numerically verified within its model, not experimentally validated.

Stop rules

Do not promote cases with failed flux balance, absorber dependence, missing channels, unresolved resonances or unsupported temperature extrapolation. Do not replace a failed point with a smooth curve. Do not infer qubit T1/T2*, single-phonon operation or measured device performance from classical elastic results. Preserve failures in the result index with the next discriminating test.

10. Campaign size and milestones

Work package I | Produce substantial results without wasting compute on unvalidated sweeps.

Milestone 1: functioning platform

Implement the configuration/schema and verification suite. Reproduce the small archived smoke case. Validate homogeneous 3D lead modes for all three thicknesses. Deliver a runnable short example, environment lock, source hashes, resource estimates and automated regression report. This milestone proves implementation readiness, not the new scientific conclusions.

Milestone 2: validated straight-device atlas

Screen 3 thicknesses x 7 proposed temperatures x 2 device types (periodic reference and defect) = 42 spectra, only where supported by the selected material policy. Homogeneous calibration is additional. At 121 initial frequency points this is 5082 frequency configurations, before multiplying by incident-channel count, convergence repeats or material scenarios. Multiple right-hand sides may share one operator factorization. Measure cost first; do not launch this count blindly.

Milestone 3: circular screening and selection

At one supported reference temperature screen 4 nonzero radii x 3 thicknesses = 12 open-bend spectra, plus matched straight controls. Select scientifically informative cases, including a non-improving case, for full temperature sweeps. Investigate closed and bus-coupled rings separately after open-bend calibration. Decide whether extra dimension interactions merit expansion using sensitivity results, not visual appeal.

Milestone 4: uncertainty and synthesis

For finalists vary justified geometry/material tolerances, tighten meshes and frequency sampling, and repeat independent controls. Deliver a comparison whose error bars resolve the conclusions. If they do not, report an unresolved ranking. Prepare manuscript-ready methods, captions and tables plus a reproducibility archive; request independent review of physics and analysis before submission.

Resource and retention policy

Pilot the largest thickness/channel case before budgeting. Record memory per factorization, wall time per frequency, field-storage size and anticipated total disk use. Set RAM, disk and elapsed-time limits in configuration; checkpoint and stop cleanly before exhausting resources. Retain compact scattering tables for every case, full fields for representative/critical points, and failure diagnostics. Never delete original evidence or launch paid/large unattended resources without corresponding authorization.

11. Required scientific results

Produce a coherent research dataset, not hundreds of redundant plots.

Core result tables

1. Material provenance and valid temperature ranges. 2. Geometry/thickness/circular-pattern matrix. 3. Mode inventory and cutoffs. 4. Polarization/velocity dispersion. 5. Full channel scattering coefficients. 6. Total T/R/absorption and energy imbalance. 7. Mesh, port and frequency convergence. 8. Temperature-dependent edge and mode shifts. 9. Circular-versus-straight matched comparison. 10. Uncertainty and sensitivity. 11. Runtime/storage benchmark. 12. Claim-by-claim validation status. Every row must map to a configuration, source version and raw result.

Figure families, produced only when supported

Dimensioned geometry and thickness views; straight/circular patterns; material-property curves; polarization-resolved dispersion; cutoff maps; representative 3D mode profiles; T/R spectra; mode-conversion matrices; frequency-thickness transmission maps; frequency-temperature maps; curvature maps; displacement/strain/energy-flux snapshots; absorber/mesh convergence; fixed-property temperature null results; material-driven thermal shifts; ring mode families; coupling/linewidth studies; uncertainty bands; design trade-offs; and compute scaling. A family may contain several useful panels, but each must answer a specific question.

Visible figure identity

At the top of every new figure, print Athul Prem and the athul_simulation_pnc_ identifier in a legible, unobtrusive line. Put scientific title, dimensions, temperature, orientation and model policy in the caption or panel labels. Use the identifier in the image filename and result index. Historical archived images remain unchanged unless deliberately re-exported as new figures. All text in the report should be black on white; scientific plots may use distinguishable colors with accessible legends.

Final report and conclusion structure

State the strongest supported findings, then explain their uncertainty and physical range. Separate elastic coherent propagation from occupancy-weighted and dissipative predictions. Include the exact number of temperatures, geometries, channels, solved frequencies and rejected/failed points, not just planned counts. Report whether circular geometry changes transmission, conversion or confinement significantly, which thicknesses support which modes, and which temperature effects are established versus assumed.

The final package must include methods sufficient to reproduce the claims, primary material references actually consulted, configurations, new _AP_pnc.py scripts, compact raw data, plotting inputs and a limitations section. No result is made publishable by formatting alone: source provenance, correct physics, independent convergence and reproducibility determine what can responsibly be claimed.